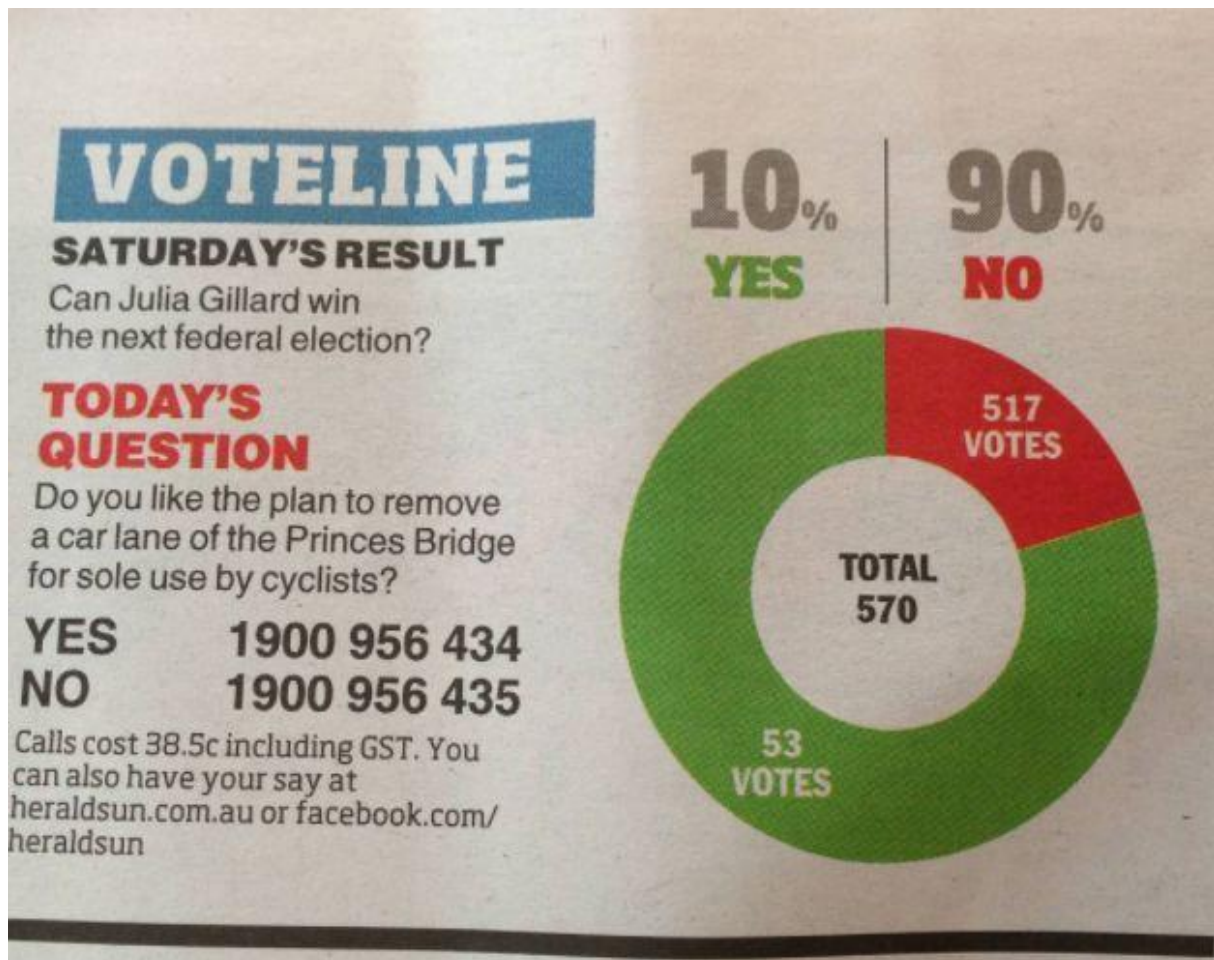




1. Bad data visualization example

The chosen bad data visualization is a donut chart published under the title “Voteline – Saturday’s Result” in an Australian newspaper.

The visualization presents the results of a public poll asking the question:

“Do you like the plan to remove a car lane of the Princes Bridge for sole use by cyclists?”

The chart displays the results as 90% “No” and 10% “Yes”, using a red and green donut chart. In the center of the chart, the total number of votes is shown as 570.

2. Problems in the visualization

This visualization contains several serious design problems that make it misleading.

First, it violates the principle of proportional ink. The donut chart visually emphasizes the large green area, creating the impression of an overwhelming consensus. However, the total number of responses is extremely small (only 570 votes). This crucial information is visually minimized and placed inside the donut, where it is easy to overlook. As a result, the perceived importance of the result is exaggerated.

Second, the use of a donut chart is inappropriate for such a small dataset. Donut charts encode values using area, which humans perceive poorly compared to length. This makes the difference between “Yes” and “No” appear more dramatic than it actually is.

Third, the visualization is contextually misleading. Participation in the poll requires making a paid phone call, which introduces a strong selection bias. However, this limitation is not clearly communicated in the visual hierarchy of the chart. Viewers may incorrectly interpret the result as representative of public opinion.

Finally, the color usage reinforces bias. The use of saturated red and green colors carries strong emotional connotations and pushes the reader toward a polarized interpretation, rather than a neutral understanding of the data.

3. How the problems can be solved

These issues could be resolved through several design improvements.

First, the sample size should be clearly emphasized, for example in the title or subtitle of the visualization. When the number of observations is very small, this information should be immediately visible.

Second, the donut chart should be replaced with a simple bar chart showing the exact vote counts. Bar charts rely on length rather than area, making comparisons more accurate and less misleading.

Third, contextual information about how the poll was conducted, including the cost of participation and potential selection bias, should be clearly stated near the visualization.

Finally, the use of neutral, less saturated colors would reduce emotional bias and help present the data more objectively.

The image used in this assignment was obtained from the WTF Visualizations (viz.wtf) website, from a post titled “Red n Green” (post date: 15 September 2021). The image itself is a newspaper clipping from the “Voteline” section of the Herald Sun. However, because the original newspaper publication date is not visible/confirmable from the available image, the clip’s original publication date is recorded as unknown for the purposes of this assignment.

<https://viz.wtf/post/662415701683978240/red-n-green>